

Chapter 3

Schur decomposition. Self-adjoint and normal matrices. Projections and congruence.

This chapter contains the core material of linear algebra. At a high level, the theoretical results in this chapter are consequences of having an inner product. At the same time, for important classes of matrices (namely, Hermitian/symmetric matrices) we will obtain very specialized results on the eigenvalues, results that are of paramount importance in applications. The material in this chapter is classic and a lot of it is covered in standard books; e.g., in [5] and [7].

3.1 Schur Decomposition

We know that $A \in \mathbb{C}^{n \times n}$ can be triangularized by similarity in $\mathbb{C}^{n \times n}$ (see Theorem 1.5.37). A very important result is that unitary similarity is sufficient to achieve the triangularization of a matrix $A \in \mathbb{C}^{n \times n}$.

Theorem 3.1.1 (Schur) *Given $A \in \mathbb{C}^{n \times n}$ with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. Then, there is a unitary U such that $U^*AU = R$, where R is upper triangular. The eigenvalues on the diagonal of R can be arranged in any desired way.*

Pf. The proof is similar to that of Theorem 1.5.37, and it proceeds by induction on n . The case $n = 1$ is obvious, so suppose that the result holds for matrices in $\mathbb{C}^{(n-1) \times (n-1)}$ and show it for $A \in \mathbb{C}^{n \times n}$.

Suppose that we want to see the eigenvalues along the diagonal of R as $\lambda_1, \lambda_2, \dots, \lambda_n$. Let v_1 be such that $Av_1 = \lambda_1 v_1$ and $v_1^* v_1 = 1$. Extend v_1 to a basis of $\mathbb{C}^n$: $v_1, y_2, \dots, y_n$. Use the Gram-Schmidt process to produce an orthonormal basis: $v_1, z_2, \dots, z_n$. Form unitary $U_1 = [v_1, z_2, \dots, z_n] \in \mathbb{C}^{n \times n}$, $U_1^* U_1 = I$. We have $U_1^* A U_1 = \begin{bmatrix} \lambda_1 & | & * \cdots * \\ \hline -- & | & -- \\ 0 & | & A_1 \end{bmatrix}$,

where $A_1 \in \mathbb{C}^{n-1, n-1}$ has eigenvalues $\lambda_2, \dots, \lambda_n$.

By induction, there is a matrix $V_2 \in \mathbb{C}^{n-1, n-1}$ unitary such that $V_2^* A_1 V_2 = \begin{bmatrix} \lambda_2 & * & \cdots & * \\ 0 & \lambda_3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}$. Form $U_2 = \begin{bmatrix} 1 & | & 0 \cdots 0 \\ \hline -- & | & -- \\ 0 & | & V_2 \end{bmatrix}$, then $U_2^* U_1^* A U_1 U_2 = \begin{bmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ 0 & \cdots & 0 & \lambda_n \end{bmatrix}$.
□

Remark 3.1.2 In general, U in Theorem 3.1.1 is not unique, nor is R . For ex-

ample, take $R_1 = \begin{bmatrix} 1 & 1 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, and $R_2 = \begin{bmatrix} 2 & -1 & 3\sqrt{2} \\ 0 & 1 & \sqrt{2} \\ 0 & 0 & 3 \end{bmatrix}$, which are unitarily equivalent

via $U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix}$, $U^* R_1 U = R_2$, and the off-diagonal entries of R_1 and R_2

are quite different! (Of course, see Theorem 2.1.12, the sum of their squares is the

same). As another example, with repeated eigenvalues, consider $R_1 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$, and

$R_2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, which are trivially unitarily equivalent via $U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Note that the example in the last Remark also shows that two matrices which are simultaneously triangularized by unitary similarity, in general do not commute. The converse, however, is true (cfr. with Exercise 1.5.41).

Exercise 3.1.3 Let $A, B \in \mathbb{C}^{n \times n}$. Show that if $AB = BA$, then A and B can be simultaneously triangularized by a unitary similarity transformation. [Hint: See Theorem 1.5.66]

Exercise 3.1.4 Let $R = \begin{bmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{bmatrix} \in \mathbb{C}^{2 \times 2}$, $\lambda_1 \neq \lambda_2$ and $b \neq 0$. Suppose that we want

the triangular form $\begin{bmatrix} \lambda_2 & c \\ 0 & \lambda_1 \end{bmatrix}$. Show constructively how we can obtain this form (i.e., swap the eigenvalues of R), with a unitary transformation of R . (You must exhibit the transformation; also, note that $|c|^2 = |b|^2$).

Next, to answer a question posed at the end of Chapter 1, let us see what happens when we have a matrix $A \in \mathbb{R}^{n \times n}$ and try to use orthogonal transformations to bring it to triangular form. The issue is how to deal with complex conjugate eigenvalues, since obviously they preclude us from being able to obtain a full triangularization maintaining the triangular factor in the real field.¹

Theorem 3.1.5 (Real Schur's Theorem: Quasi-Upper Triangular Form) *Let $A \in \mathbb{R}^{n \times n}$. Then, there exists $Q \in \mathbb{R}^{n \times n}$, orthogonal, such that $Q^T A Q$ is in quasi-upper triangular form, that is:*

$$Q^T A Q = R = \begin{bmatrix} R_1 & * & \cdots & * \\ & R_2 & \ddots & \vdots \\ & & \ddots & * \\ & & & R_k \end{bmatrix}, \quad \text{for some } k \leq n,$$

where each R_i is either $(1, 1)$ (hence, it is a real eigenvalue), or it is $(2, 2)$ containing a pair of complex conjugate eigenvalues. Eigenvalues can be ordered, that is the R_i 's can, in any desired way, except that complex conjugate eigenvalues must be in the $(2, 2)$ blocks.

Proof. Whereas the real eigenvalues can be treated as in the previous Schur's Theorem 3.1.1, let's see how to deal with a complex conjugate pair. So, suppose $\lambda, \bar{\lambda} = \alpha \pm i\beta$, $\beta \neq 0$, and let $x \neq 0$ be an eigenvector associated to λ : $Ax = \lambda x$. Write $x \in \mathbb{C}^n$ as $x = u + iv$, $u, v \in \mathbb{R}^n$, and notice that also $A\bar{x} = \bar{\lambda}\bar{x}$. Therefore, we have $A[u, v] = [u, v] \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix}$, all real. Since $\lambda \neq \bar{\lambda}$, then $\{x, \bar{x}\}$ are linearly independent

¹Naturally, we could ask the same question relative to the Jordan form. In this case, one obtains the so called *real Jordan form*, which is seldom used in practice: "Given $A \in \mathbb{R}^{n \times n}$, there exists invertible $V \in \mathbb{R}^{n \times n}$, such that $V^{-1}AV = J$, where $J = \text{diag}(J_k, k = 1, \dots, p)$, and $J_k = \begin{bmatrix} \lambda_k & 1 & \cdots & 0 \\ & \ddots & \ddots & \vdots \\ & & \lambda_k & 1 \\ & & & \lambda_k \end{bmatrix}$, if $\lambda_k \in \mathbb{R}$ (as in the general case). Instead, if $\lambda_k = \alpha_k + i\beta_k$, and $\bar{\lambda}_k = \alpha_k - i\beta_k$, is a pair of complex conjugate eigenvalues, with $\beta_k \neq 0$, then the diagonal blocks J_k have the form $J_k = \begin{bmatrix} D_k & I_2 & & \\ & \ddots & \ddots & \\ & & D_k & I_2 \\ & & & D_k \end{bmatrix}$, where $D_k = \begin{bmatrix} \alpha_k & \beta_k \\ -\beta_k & \alpha_k \end{bmatrix}$."

over $\mathbb{C}$, and so $\{u, v\}$ are linearly independent over $\mathbb{R}$. Now extend $\{u, v\}$ to a basis for $\mathbb{R}^n$, and use the Gram-Schmidt process on this to obtain an orthonormal basis of $\mathbb{R}^n$. Form the orthogonal matrix Q by using the orthonormal basis just found as Q 's columns. Note that the first two columns of Q , q_1, q_2 , span the same subspace as the pair u, v , and indeed $\begin{bmatrix} u & v \end{bmatrix} = \begin{bmatrix} q_1 & q_2 \end{bmatrix} R_{11}$, where $R_{11} \in \mathbb{R}^{2 \times 2}$ and it is triangular and invertible. Therefore, $Q^T A Q = \begin{bmatrix} R_1 & | & * \\ \hline -- & | & -- \\ 0 & | & A_1 \end{bmatrix}$, where $R_1 = R_{11} \begin{bmatrix} \alpha & \beta \\ -\beta & \alpha \end{bmatrix} R_{11}^{-1}$ has eigenvalue $\lambda, \bar{\lambda}$, and $A_1 \in \mathbb{R}^{n-2, n-2}$. Now, consider A_1 and proceed inductively. $\square$

Exercise 3.1.6 Consider the real matrix $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ with complex conjugate eigenvalues $\pm i$. Because of Theorem 2.1.12, we know that it is not possible to find Q orthogonal, such that $Q^T A Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Nevertheless, we can say more about the structure of the $(2, 2)$ blocks occurring along the diagonal of R in Theorem 3.1.5. Show that, relative to a $(2, 2)$ block B corresponding to the complex conjugate pair $\alpha \pm i\beta$, $\beta \neq 0$, we can always find orthogonal $Q \in \mathbb{R}^{2 \times 2}$ such that $Q^T B Q = \begin{bmatrix} \alpha & \gamma\beta \\ -\frac{\beta}{\gamma} & \alpha \end{bmatrix}$, for some $\gamma \neq 0$. What is γ for $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$? Is it unique?

Exercise 3.1.7 (On the degree of uniqueness of Schur forms, part 1) Here we explore the degree to which unitary (respectively, orthogonal) similarities to upper triangular form (respectively, quasi-upper triangular form) are unique.

- (a) Suppose that all eigenvalues $\lambda_1, \dots, \lambda_n$, of $A \in \mathbb{C}^{n \times n}$ are distinct, and that we have transformed A , by unitary similarity, to upper triangular form with the eigenvalues appearing on the diagonal in the order $\lambda_1, \dots, \lambda_n$. Discuss the degree of uniqueness of the unitary transformation. In other words, is the unitary transformation unique? If not, what freedom do you have in choosing U ?
- (b) Suppose that all eigenvalues $\lambda_1, \dots, \lambda_n$, of $A \in \mathbb{R}^{n \times n}$ are distinct (though not necessarily all of them real), and that we transform A using orthogonal similarity to quasi-upper triangular form with a given ordering of the eigenvalues (complex conjugate eigenvalues appear on a $(2, 2)$ diagonal block but are not otherwise distinguished). Discuss uniqueness of the orthogonal factor.

(c) [Harder] **Briefly** discuss –for cases (a) and (b)– what we can say if the eigenvalues are not distinct.

[Hint. See Exercise 1.5.46. Alternatively, for (a), let U and V be such that $U^*AU = R$ and $V^*AV = S$ are both upper triangular and let $H = U^*V$. What equation does H satisfy?]

An important consequence of Schur's Theorem 3.1.1 (complex case) is that “the set of diagonalizable matrices is dense in the set of matrices.” This means that given any non-diagonalizable matrix, there is a matrix arbitrarily close to it which is diagonalizable. We will actually show a stronger result, that, given any matrix, there is a matrix arbitrarily close to it and with distinct eigenvalues.

Theorem 3.1.8 *Given $A \in \mathbb{C}^{n \times n}$. Then, $\forall \varepsilon > 0$, there exist $A(\varepsilon) \in \mathbb{C}^{n \times n}$ with distinct eigenvalues and such that*

$$\|A - A(\varepsilon)\|_F^2 < \varepsilon.$$

Pf. Let U , unitary, be such that $U^*AU = R$ is upper triangular with eigenvalues $\lambda_1, \dots, \lambda_n$, along the diagonal. Let $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{C}$ such that $|\alpha_i| < \left(\frac{\varepsilon}{n}\right)^{1/2}$ and such that $\lambda_1 + \alpha_1, \lambda_2 + \alpha_2, \dots, \lambda_n + \alpha_n$ are all distinct. Take $E(\varepsilon) = \text{diag}(\alpha_1, \dots, \alpha_n)$ and form $R(\varepsilon) = R + E(\varepsilon)$ and, from this, form $A(\varepsilon) = UR(\varepsilon)U^*$. Now, clearly $A(\varepsilon)$ has distinct eigenvalues, and further observe that

$$\|A(\varepsilon) - A\|_F^2 = \|R(\varepsilon) - R\|_F^2 = \sum_{i,j} |R_{ij}(\varepsilon) - R_{ij}|^2 = \sum_{i=1}^n |\alpha_i|^2 < \varepsilon. \quad \square$$

Exercise 3.1.9 *State and prove an analog of Theorem 3.1.8 for $A \in \mathbb{R}^{n \times n}$, making $A(\varepsilon) \in \mathbb{R}^{n \times n}$. [Hint: Produce a real matrix with distinct eigenvalues, whether real or complex conjugate.]*

In light of Theorem 3.1.8, and Exercise 3.1.9, it is natural to ask why we should bother with non-diagonalizable matrices. There are at least two reasons. The first is that, when computing eigenvalues, it is very difficult to distinguish between equal or merely close eigenvalues. The second reason is the content of the following example.

Example 3.1.10 *Think about this example. Suppose we have the real valued function $A(t) = \begin{pmatrix} 0 & 1 \\ t & 0 \end{pmatrix}$, for $t \in [-1, 1]$, which is clearly non-diagonalizable at $t = 0$. We*

claim that if we take the real matrix $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and perturb the function $A(t)$, $t \in [-1, 1]$, to $A(t) + \varepsilon B$, with ε small, then this perturbed function will be non-diagonalizable at some point in a neighborhood of $t = 0$. In other words, whereas each matrix $A(t)$, for given t , may be effectively perturbed into a diagonalizable one, the entire family cannot. (Incidentally, the choice of B above is for simplicity only; any choice of B so that εB is small would do).

Exercise 3.1.11 Verify the claims in Example 3.1.10.

3.2 Self-adjoint (symmetric) and normal matrices

In general, in Schur's Theorem 3.1.1, U and R are not uniquely determined. However, the quantity

$$\nu = \sum_{i < j} |R_{ij}|^2 \quad (3.2.1)$$

is always uniquely determined. (This is simply a consequence of Frobenius norm invariance, see Theorem 2.1.12.) For reasons that will be clarified below, ν is called “departure from normality” of A . Schur's theorem, and the fact that ν is invariant under unitary transformations, give us an important consequence: In general, we cannot expect being able to transform a matrix to diagonal form with unitary similarity. Or, to put it in other words, we are able to do so only if—during the process of triangularizing it—we have effectively diagonalized it!

For this reason, our next task is to specialize Schur's theorem to very important classes of matrices, those which are diagonalizable by a unitary transformation.

Definition 3.2.1 A matrix $A \in \mathbb{C}^{n \times n}$ such that $A^* = A$ is called Hermitian (or self-adjoint). If $A^* = -A$, then A is called anti-Hermitian (or skew-Hermitian). In the real case, $A \in \mathbb{R}^{n \times n}$, if $A^T = A$ then A is called symmetric (or, again, self-adjoint), whereas if $A^T = -A$ then A is called anti-symmetric (or skew-symmetric).

Unless otherwise explicitly stated, we will use the term symmetric just for real matrices; if needed, we will use the term “complex symmetric” to identify a complex matrix A which satisfies $A = A^T$.

Remark 3.2.2 If $A^* = -A$, then iA is Hermitian (similarly, if $A^* = A$ then iA is anti-Hermitian). Therefore, a result proven for Hermitian matrices has an immediate counterpart for anti-Hermitian matrices.

Hermitian (symmetric) matrices appear pervasively in applications, and this justifies paying special attention to these matrices. In particular, symmetric (Hermitian) matrices are of fundamental importance in studying behavior of functions near critical points.

Example 3.2.3 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be sufficiently smooth (at least twice continuously differentiable). Consider the Taylor expansion of f near a point x_0 : $f(x) = f(x_0) + (\nabla f(x_0))^T(x - x_0) + \frac{1}{2}Q(x - x_0) + r(\|x - x_0\|)$ (here, $\frac{1}{\|x - x_0\|^2}r(\|x - x_0\|) \rightarrow 0$ as $\|x - x_0\| \rightarrow 0$). The function $Q(x - x_0)$ is the quadratic function given by

$$Q(y) = \sum_{i,j} H_{ij} y_i y_j, \quad \text{and} \quad H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j} \Big|_{x=x_0}.$$

Therefore, $Q(x - x_0) = (x - x_0)^T H (x - x_0)$, where H is the Hessian, and is symmetric: $H_{ij} = H_{ji}$.

Now, if x_0 is a critical point $\Rightarrow \nabla f(x_0) = 0$ and thus the quadratic function determines the nature of the critical point. E.g., one of the most important results in unconstrained optimization guarantees that a critical point is a (strict) minimum if the Hessian has all positive eigenvalues; see Exercise 3.2.46-(8). $\square$

The example below sheds light into the nature of the eigenvalues of Hermitian (anti-Hermitian) matrices.

Example 3.2.4 Let $A \in \mathbb{C}^{n \times n}$. Define $A_s = (A + A^*)/2$ (Hermitian, or self-adjoint, part of A) and $A_a = (A - A^*)/2$ (anti-Hermitian part of A). Clearly, $A = A_s + A_a$ and this is the unique decomposition of A into the sum of a Hermitian and an anti-Hermitian matrix. Then, we claim that:

$$\begin{aligned} \operatorname{Re}\langle x, Ax \rangle &= \langle x, A_s x \rangle \\ \operatorname{Im}\langle x, Ax \rangle &= \langle x, A_a x \rangle. \end{aligned}$$

The verification of this claim is simple. Take

$$\begin{aligned} \langle x, Ax \rangle &= x^* A^* x = \frac{x^* (A^* - A + A + A^*) x}{2} \\ &= \frac{x^* (A^* + A) x}{2} + \frac{x^* (A^* - A) x}{2} = \langle x, A_s x \rangle + \langle x, A_a x \rangle. \end{aligned}$$

Now:

$$\langle x, A_s x \rangle = (\langle x, A_s x \rangle)^* \therefore \langle x, A_s x \rangle \in \mathbb{R}$$

$$\text{and } \langle x, A_a x \rangle = -(\langle x, A_a x \rangle)^* \therefore \langle x, A_a x \rangle \in i\mathbb{R} ,$$

as claimed.

Furthermore, if (λ, x) is an eigenpair of A , $Ax = \lambda x$, such that $x^*x = 1$, then

$$\begin{cases} x^*Ax = \lambda \\ x^*A^*x = \bar{\lambda} \end{cases}, \text{ from which } \operatorname{Re}\lambda = x^*A_s x, \quad \text{and} \quad \operatorname{Im}\lambda = x^*A_a x. \quad \square$$

An immediate consequence of the above Example is the following result.

Corollary 3.2.5 *If $A \in \mathbb{C}^{n \times n}$ is Hermitian, respectively $A \in \mathbb{R}^{n \times n}$ is symmetric, then its eigenvalues are real. If $A \in \mathbb{C}^{n \times n}$ is anti-Hermitian, respectively $A \in \mathbb{R}^{n \times n}$ is anti-symmetric, then its eigenvalues are purely imaginary.* $\square$

We are now ready for one of the most often used results in linear algebra.

Theorem 3.2.6 (Spectral Theorem for Hermitian matrices) *Let $A \in \mathbb{C}^{n \times n}$ be Hermitian: $A = A^*$. Then, A is unitarily diagonalizable. That is, there exists unitary $U \in \mathbb{C}^{n \times n}$ such that $D = U^*AU$ is diagonal. The eigenvalues can be ordered in any desired way along the diagonal of D .*

Pf. From Schur's Theorem 3.1.1, we have that there exists U unitary such that $U^*AU = R$, R upper triangular with eigenvalues ordered as we like. Therefore, $U^*A^*U = R^* \Rightarrow R = R^* \therefore R_{ij} = 0, i \neq j$, and thus R is diagonal (and the diagonal is real). $\square$

Much the same result holds if $A \in \mathbb{R}^{n \times n}$. Before stating the theorem in this case, let us make an observation.

Observation 3.2.7 *If $A \in \mathbb{R}^{n \times n}$ with eigenvalue $\lambda \in \mathbb{R}$ (e.g., if $A = A^T$), then we can (and will) take the corresponding eigenvector to be real. This is simply because if $Ax = \lambda x$, $x \neq 0$ and we let $x = u + iv$, with $u, v \in \mathbb{R}^n$, then we must have*

$$Au + iAv = \lambda u + i\lambda v \text{ and so } \begin{cases} Au = \lambda u \\ Av = \lambda v \end{cases}. \text{ So, we can take } v = 0 \text{ and } x = u \text{ to be a real eigenvector.}$$

Theorem 3.2.8 (Spectral theorem for symmetric matrices) *Let $A \in \mathbb{R}^{n \times n}$ be symmetric: $A = A^T$. Then, there exists orthogonal $Q \in \mathbb{R}^{n \times n}$ such that $D = Q^T A Q$ is diagonal. The eigenvalues can be ordered in any desired way along the diagonal of D .*

Pf. Since all eigenvalues are real, the Real Schur's Theorem 3.1.5 gives that there exists Q , orthogonal, such that $Q^T A Q = R$, with R upper triangular with eigenvalues ordered as we like. But then $Q^T A^T Q = R^T \Rightarrow R$ is diagonal. $\square$

Exercise 3.2.9 (On the degree of uniqueness of Schur forms, part 2) *Here we continue (see Exercise 3.1.7) exploring the degree of uniqueness of unitary (respectively, orthogonal) similarity of a Hermitian (respectively, symmetric) matrix to diagonal form.*

We will assume that the eigenvalues appear ordered along the diagonal: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Further, we let $\mu_1 > \mu_2 > \dots > \mu_p$ be the distinct eigenvalues of A , of respective algebraic multiplicities $n_1, \dots, n_p$.

- (a) Let $A = A^* \in \mathbb{C}^{n \times n}$ and U be a unitary matrix having achieved the ordered diagonalization of A as above. Show that U is unique up to a “unitary block-diagonal” transformation. That is, show that all unitary transformations achieving the above ordered diagonalization must be of the type:

$$V = U \begin{bmatrix} Z_1 & & \\ & \ddots & \\ & & Z_p \end{bmatrix}, \text{ where } Z_i \in \mathbb{C}^{n_i \times n_i}, \quad Z_i^* Z_i = I_{n_i}, \quad i = 1, \dots, p. \quad (3.2.2)$$

- (b) Repeat part (a) for the case of $A = A^T \in \mathbb{R}^{n \times n}$ and the transformation to diagonal form being orthogonal.

Exercise 3.2.10 Suppose that $A = A^* \in \mathbb{C}^{n \times n}$ and $B = B^* \in \mathbb{C}^{n \times n}$, and $AB = BA$. From Theorem 1.5.31, we know that A and B can be simultaneously diagonalized. Show that the diagonalizing transformation can be taken unitary. Repeat for the case of symmetric $A, B \in \mathbb{R}^{n \times n}$ and orthogonal transformation.

Remark 3.2.11 We have achieved diagonalization for a Hermitian (respectively, symmetric) matrix without the need to assume distinct eigenvalues. Hermitian (symmetric) matrices lead to a basis of orthonormal eigenvectors: Hermitian matrices are diagonalized by a unitary similarity transformation. So, conceptually at least, we can think of a given Hermitian (symmetric) matrix as being diagonal. For example, in Example 3.2.3, we can change coordinates and diagonalize the Hessian with an orthonormal basis; if we do so, the orthonormal vectors are called principal axes and they provide the new coordinate system.

- Finally, let us see how far we can push diagonalization by unitary similarity. To motivate the next definition, consider the following.

Take $A \in \mathbb{C}^{n \times n}$ and let $A = A_s + A_a$ be its unique decomposition in Hermitian and anti-Hermitian components. Since $A_s = A_s^*$, there exists U unitary such that $U^* A_s U = D$ (and real). Likewise (since iA_a is Hermitian) also A_a can be unitarily diagonalized. We also know (see Theorem 1.5.31) that two diagonalizable matrices are simultaneously diagonalizable if and only if they commute. Now, observe the following:

$$A_s A_a = \frac{A + A^*}{2} \frac{A - A^*}{2} = \frac{A^2 + A^* A - A A^* - (A^*)^2}{4}$$

$$A_a A_s = \frac{A - A^*}{2} \frac{A + A^*}{2} = \frac{A^2 - A^* A + A A^* - (A^*)^2}{4}$$

$$\therefore A_s A_a = A_a A_s \Leftrightarrow A A^* - A^* A = A^* A - A A^* \Leftrightarrow A A^* = A^* A.$$

Definition 3.2.12 If $A \in \mathbb{C}^{n \times n}$ (respectively, $A \in \mathbb{R}^{n \times n}$) is such that $A A^* = A^* A$ (respectively, $A A^T = A^T A$) then A is called a *normal matrix*.

Therefore, we know that normal matrices are simultaneously diagonalizable, since both A_s and A_a are so, and $A_s A_a = A_a A_s$. What we will show next is that the diagonalizing transformation can be taken to be unitary. The following observation is simple to verify, and it will come in handy.

Observation 3.2.13 Unitary similarity of a normal $A \in \mathbb{C}^{n \times n}$ preserves normality. Similarly, if $A \in \mathbb{R}^{n \times n}$ is normal and Q is orthogonal, then $Q^T A Q$ is normal.

Theorem 3.2.14 (Spectral Theorem for normal matrices) $A \in \mathbb{C}^{n \times n}$ is normal $\Leftrightarrow$ it is unitarily diagonalizable.

Pf. ($\Leftarrow$) Let U , unitary, be such that $U^* A U = D$, where D is diagonal. Thus, $U^* A^* U = D^* \Rightarrow A A^* = U (U^* A U U^* A^* U) U^* = U D D^* U^* = U D^* D U^* = U D^* U^* U D U^* = A^* A$.

($\Rightarrow$) We show that if a matrix is normal and upper triangular, then it must be diagonal. Proceed by induction on n . Obviously the result is true for $n = 1$.

Next, suppose that the result is true for normal matrices in $\mathbb{C}^{(n-1) \times (n-1)}$ and show it for $A \in \mathbb{C}^{n \times n}$.

Let unitary $U \in \mathbb{C}^{n \times n}$ be such that $U^* A U = R$ is upper triangular (Schur's theorem). Then, since $A^* A = A A^*$, then, because of Observation 3.2.13, also $R R^* = R^* R$.

Write $R = \begin{bmatrix} r_{11} & a^* \\ 0 & R_1 \end{bmatrix} \Rightarrow R^*R = \begin{bmatrix} r_{11}^* & 0 \\ a & R_1^* \end{bmatrix} \begin{bmatrix} r_{11} & a^* \\ 0 & R_1 \end{bmatrix} = \begin{bmatrix} r_{11}^*r_{11} & r_{11}^*a^* \\ ar_{11} & R_1^*R_1 + aa^* \end{bmatrix}$ and $RR^* = \begin{bmatrix} r_{11} & a^* \\ 0 & R_1 \end{bmatrix} \begin{bmatrix} r_{11}^* & 0 \\ a & R_1^* \end{bmatrix} = \begin{bmatrix} r_{11}r_{11}^* + a^*a & a^*R_1^* \\ R_1a & R_1R_1^* \end{bmatrix}$. Thus, from $r_{11}^*r_{11} = r_{11}r_{11}^* + a^*a$ we immediately get $a = 0 \therefore R_1R_1^* = R_1^*R_1$ and so R_1 is normal. Now use the induction hypothesis. $\square$

Remark 3.2.15 *Careful: Theorem 3.2.14 says that there exists U , unitary, diagonalizing a normal matrix A . But it does not say that a unitary similarity transformation diagonalizing A_s also diagonalizes A_a .*

For example, take $A \in \mathbb{C}^{n,n}$, $A = \begin{bmatrix} C & 0 \\ 0 & B \end{bmatrix}$, $C^* = C \in \mathbb{C}^{n_1,n_1}$ and $B^* = -B \in \mathbb{C}^{n_2,n_2}$, $n_1 + n_2 = n$, so that A is normal. Here, $A_s = \begin{bmatrix} C & 0 \\ 0 & 0 \end{bmatrix}$, $A_a = \begin{bmatrix} 0 & 0 \\ 0 & B \end{bmatrix}$. Let $U_1 \in \mathbb{C}^{n_1,n_1}$, $U_1^*U_1 = I$, be such that $U_1^*CU_1 = D_C$ (diagonal and real), and likewise let $U_2 \in \mathbb{C}^{n_2,n_2}$, $U_2^*U_2 = I$, be such that $U_2^*BU_2 = D_B$ (diagonal and purely imaginary); then, $U = \begin{pmatrix} U_1 & 0 \\ 0 & U_2 \end{pmatrix}$ is unitary and diagonalizes each of A_s, A_a, A . But, we could have also taken

$$V_1 = \begin{pmatrix} U_1 & 0 \\ 0 & \widehat{U} \end{pmatrix}, \quad V_1 \text{ unitary} : V_1^*A_sV_1 = \begin{pmatrix} D_C & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow V_1^*AV_1 = \begin{pmatrix} D_C & 0 \\ 0 & \widehat{U}^*B\widehat{U} \end{pmatrix}$$

and generally $\widehat{U}^*B\widehat{U}$ is not diagonal.

However, if the orthonormal eigenvectors of A_s and A_a are the same, then A is normal and it is diagonalized by these very same eigenvectors. $\square$

Exercise 3.2.16 *Show that if the orthonormal eigenvectors of A_s are also eigenvectors of $A_a \Rightarrow A = A_s + A_a$ is normal.*

In the real case, similar results hold. To begin with, just as in the complex case, being normal is equivalent to $A_sA_a = A_aA_s$, where $A_s = \frac{A+A^T}{2}$, $A_a = \frac{A-A^T}{2}$, that is $A^TA = AA^T$. Moreover, see Observation 3.2.13, normality is preserved by orthogonal similarity. The spectral decomposition result now reads as follows.

Theorem 3.2.17 (Spectral Theorem for real normal matrices) *Let $A \in \mathbb{R}^{n \times n}$. Then, A is normal if and only if there exists $Q \in \mathbb{R}^{n \times n}$, orthogonal, such that*

$$Q^T A Q = E = \begin{bmatrix} D_1 & & & \\ & D_2 & & \\ & & \ddots & \\ & & & D_p \end{bmatrix}, \quad (3.2.3)$$

where each D_j is either a $(1,1)$ block (real eigenvalues) or a $(2,2)$ block of the type $\begin{pmatrix} \alpha_j & \beta_j \\ -\beta_j & \alpha_j \end{pmatrix}$, $\beta_j \neq 0$ (complex conjugate eigenvalues).

Pf. $(\Leftarrow) Q^T A Q = E \Rightarrow Q^T A^T Q = E^T \Rightarrow A = Q E Q^T$, $A^T = Q E^T Q^T \Rightarrow A A^T = Q E Q^T Q E^T Q^T = Q E E^T Q^T = Q E^T E Q^T = Q E^T Q^T Q E Q^T = A^T A$.

$$(\text{Note: } \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \begin{bmatrix} a & -b \\ b & a \end{bmatrix} = \begin{bmatrix} a^2 + b^2 & 0 \\ 0 & a^2 + b^2 \end{bmatrix} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \begin{bmatrix} a & b \\ -b & a \end{bmatrix}).$$

$(\Rightarrow)$ Using the Real Schur's Theorem 3.1.5, and Observation 3.2.13, we can assume that A is normal and in quasi-triangular form as

$$A = \begin{bmatrix} R_{11} & A_{12} & \cdots & A_{1p} \\ 0 & R_{22} & \ddots & \vdots \\ \vdots & \ddots & \ddots & A_{p-1,p} \\ 0 & \cdots & 0 & R_{pp} \end{bmatrix},$$

where R_{11} is upper triangular and has all the real eigenvalues of A , and $R_{22}, \dots, R_{pp}$ are $(2,2)$ real matrices corresponding to complex conjugate eigenvalues. Say $R_{11} \in \mathbb{R}^{n_1, n_1}$, so that $A_{1k} \in \mathbb{R}^{n_1, 2}$, $k = 2, \dots, p$. Also, $A_{ij} \in \mathbb{R}^{2 \times 2}$, $i = 2, \dots, p-1$, and $j = i+1, \dots, p$.

Now, since $A^T A = A A^T$, then $R_{11}^T R_{11} = R_{11} R_{11}^T + A_{12} A_{12}^T + \cdots + A_{1p} A_{1p}^T$ and therefore also $\text{tr}(R_{11}^T R_{11}) = \text{tr}(R_{11} R_{11}^T + \cdots + A_{1p} A_{1p}^T) = \text{tr}(R_{11} R_{11}^T) + \text{tr}(A_{12} A_{12}^T) + \cdots + \text{tr}(A_{1p} A_{1p}^T)$. But² $\text{tr}(R_{11} R_{11}^T) = \text{tr}(R_{11}^T R_{11})$, and so we must have

$$\text{tr}(A_{12} A_{12}^T) + \cdots + \text{tr}(A_{1p} A_{1p}^T) = \sum_{i=1}^{n_1} \sum_{j=1}^2 (A_{12})_{ij}^2 + \cdots + \sum_{i=1}^{n_1} \sum_{j=1}^2 (A_{1p})_{ij}^2 = 0$$

²It is easy to verify by directly multiplying the matrices that $\text{tr}(CD) = \text{tr}(DC)$ for any $D, C \in \mathbb{R}^{m \times m}$; alternatively, see Corollary 5.1.11

from which it follows that $A_{12} = 0, \dots, A_{1p} = 0$. So, A really has the form

$$A = \begin{bmatrix} R_{11} & 0 & \cdots & \cdots & 0 \\ 0 & R_{22} & A_{22} & \cdots & A_{2p} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & R_{p-1,p-1} & A_{p-1,p} \\ 0 & \cdots & 0 & 0 & R_{pp} \end{bmatrix}.$$

Now, reasoning as in the proof of Theorem 3.2.14, since R_{11} is upper triangular and normal, then it must be diagonal, call it D_1 .

Moreover, since $A^T A = A A^T$, then we must also have $R_{22}^T R_{22} = R_{22} R_{22}^T + A_{23} A_{23}^T + \cdots + A_{2p} A_{2p}^T$ and this gives $\text{tr}(R_{22}^T R_{22}) = \text{tr}(R_{22} R_{22}^T) + \text{tr}(A_{23} A_{23}^T) + \cdots + \text{tr}(A_{2p} A_{2p}^T)$, from which as above we get $A_{23} = 0, \dots, A_{2p} = 0$, and R_{22} is normal.

Continuing this way, we end up with

$$A = \begin{bmatrix} D_1 & & & \\ & R_{22} & & \\ & & \ddots & \\ & & & R_{pp} \end{bmatrix},$$

where each R_{jj} is $(2, 2)$, normal, with complex conjugate eigenvalues. So, we now need to verify that a real matrix $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with complex conjugate eigenvalues $\alpha \pm i\beta$, $\beta \neq 0$, and normal (i.e., $B^T B = B B^T$), must have the form $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$, and $a = \alpha$, $b = \beta$ (or $b = -\beta$). We have

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} a^2 + b^2 & ac + bd \\ ac + bd & c^2 + d^2 \end{pmatrix}$$

$$\therefore \begin{cases} a^2 + c^2 = a^2 + b^2 \\ d^2 + b^2 = d^2 + c^2 \rightarrow c^2 = b^2 \Rightarrow b = c \text{ or } b = -c \\ ab + cd = ac + bd \rightarrow a(b - c) = d(b - c) \end{cases}.$$

Now, if $b = c$, then B would be symmetric and it would have real eigenvalues, which is excluded. So, it must be $b = -c \Rightarrow 2ba = 2bd \Rightarrow a = d$ ($b \neq 0$ otherwise again B would be symmetric) and so $B = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$ and $2a = 2\alpha$ and $a^2 + b^2 = \alpha^2 + \beta^2$ from which $a = \alpha$ and $b = \pm\beta$, as stated. $\square$

Remark 3.2.18 In Theorem 3.2.3, it is worth stressing once more the specific form of a real normal $(2, 2)$ block of complex conjugate eigenvalues $\alpha \pm i\beta$. This is a peculiarity of being normal. It is also worth emphasizing that the form (3.2.3) is obtained automatically for A normal while seeking a real Schur form of A .

- What is “normal”? For example, a diagonal matrix, or a Hermitian matrix, or an anti-Hermitian one, or a unitary matrix, or the direct sum of any of these matrices is normal. In the appropriate unitary basis, these types of matrices are all diagonal. In the real case, see Theorem 3.2.3, one can only achieve quasi-diagonal structure. For example, the next result can be immediately obtained as a consequence of Theorem 3.2.3.

Corollary 3.2.19 (Spectral Theorem for antisymmetric matrices) Let $A \in \mathbb{R}^{n \times n}$, $A^T = -A$. Then, there exists $Q \in \mathbb{R}^{n \times n}$, orthogonal, such that

$$Q^T A Q = E = \begin{bmatrix} 0_{n_1} & & & \\ & D_2 & & \\ & & \ddots & \\ & & & D_p \end{bmatrix},$$

where each D_j , $j = 2, \dots, p$, is a $(2, 2)$ block of the type $\begin{pmatrix} 0 & \beta_j \\ -\beta_j & 0 \end{pmatrix}$. □

Exercises 3.2.20

- (1) Show that $A \in \mathbb{C}^{n \times n}$ is normal $\Leftrightarrow \|Ax\|_2 = \|A^*x\|_2$, $\forall x \in \mathbb{C}^n$.
- (2) Show that $A \in \mathbb{C}^{n \times n}$ is normal $\Leftrightarrow \exists U$, unitary: $A^* = AU$.
- (3) Let $A \in \mathbb{C}^{n \times n}$, and $z \in \mathbb{C}$ be given. Show that A is normal $\Leftrightarrow A + zI$ is normal.
- (4) Show that if A is normal and $q(t)$ is a polynomial $\Rightarrow q(A)$ is normal.
- (5) Show that if $A \in \mathbb{C}^{n \times n}$ is normal, $\mathcal{N}(A) = \mathcal{N}(A^*)$ (here, $\mathcal{N}(A)$ is the null space of A). (This is trivial after the next exercise, so do it without using it.)
- (6) Show that $A \in \mathbb{C}^{n \times n}$ is normal $\Leftrightarrow$ every eigenvector of A is also eigenvector of A^* . (Cfr. with Exercise 3.2.16).
- (7) Show that if $A \in \mathbb{C}^{n \times n}$ is normal $\Rightarrow \|A\|_2 = \max_j |\lambda_j(A)|$, where $\lambda_j(A)$ are the eigenvalues of A . Is the converse also true?
- (8) Show that $\|A^*A\|_2 = \max_j (\lambda_j(A^*A))$. [Hint: Observe that the eigenvalues of A^*A are ≥ 0 , then diagonalize A^*A with a unitary transformation.]

Theorem 3.2.21 (Spectral norm of matrices) Let $A \in \mathbb{C}^{m \times n}$ (or $\mathbb{R}^{m \times n}$). Then

$$\|A\|_2 = \max_j (\lambda_j(A^*A))^{1/2}.$$

Pf. Recall that $\|A\|_2 = \max_{x: \|x\|_2=1} \|Ax\|_2$. Let x be an arbitrary unit vector, $\|x\|_2 = 1$. Then, we have

$$\|Ax\|_2^2 = \langle Ax, Ax \rangle = x^* A^* A x = \langle A^* A x, x \rangle \stackrel{\text{Cauchy-Schwartz}}{\leq} \|x\|_2 \|A^* A x\|_2$$

so $\|Ax\|_2^2 \leq \|A^* A x\|_2$ and then $\|A\|_2^2 \leq \|A^* A\|_2 = \max_j (\lambda_j(A^* A)) =: \lambda_{\max}(A^* A)$, where we used Exercise 3.2.20-(8).

Now, let $x_{\max}$ be an eigenvector (of norm 1, $\|x_{\max}\|_2 = 1$) such that $A^* A x_{\max} = \lambda_{\max}(A^* A) x_{\max}$. Then, $\langle A^* A x_{\max}, x_{\max} \rangle = \lambda_{\max}(A^* A) = \|A x_{\max}\|_2^2$. $\square$

Exercises 3.2.22

- (1) Given $A \in \mathbb{C}^{n \times n}$, normal, characterize $\text{rank}(A)$, $\text{nullity}(A)$, $\mathcal{N}(A)$, $\mathcal{R}(A)$ in terms of A 's eigenvalues/eigenvectors. If A is not normal, relate its rank to A 's eigenstructure via its Jordan form. (Here, $\mathcal{N}(A)$ and $\mathcal{R}(A)$ are the null-space (or kernel) and column-space (or range, or image) of A).
- (2) Show that $\|A\|_2 = \|A^*\|_2$ for any $A \in \mathbb{C}^{n \times n}$.
- (3) Let $A \in \mathbb{C}^{n \times n}$, and λ_j , $j = 1, \dots, n$, be its eigenvalues. Show that $\|A\|_2 \geq \max_j |\lambda_j|$, and in fact show that $\|A\| \geq \max_j |\lambda_j|$ for any consistent matrix norm $\|\cdot\|$.
- (4) Show that $\|A\|_2 \geq \frac{1}{\sqrt{n}} \|A\|_F$. Together with Exercise 2.1.14, this gives

$$\frac{1}{\sqrt{n}} \|A\|_F \leq \|A\|_2 \leq \|A\|_F.$$

3.2.1 Projections

Let us now look at some interesting implications of the spectral theorems for Hermitian (symmetric) matrices (Theorems 3.2.6 and 3.2.8).

What these theorems imply is that $\mathbb{C}^n$ (or $\mathbb{R}^n$) can be decomposed as the direct sum of pairwise orthogonal eigenspaces of a Hermitian (symmetric) matrix A . That is, if A has distinct eigenvalues $\lambda_1, \dots, \lambda_p$, of multiplicity $n_1, \dots, n_p$, respectively, then:

$$\mathbb{C}^n = V^{(1)} \oplus V^{(2)} \oplus \dots \oplus V^{(p)} \quad (\text{or } \mathbb{R}^n = V^{(1)} \oplus \dots \oplus V^{(p)})$$

where, for each $j = 1, \dots, p$, $V^{(j)}$ is the subspace spanned by the n_j eigenvectors relative to the eigenvalue λ_j (and $\lambda_j \neq \lambda_i$, $i \neq j$, $i, j = 1, \dots, p$).

Therefore, every given $v \in \mathbb{C}^n$ (or in $\mathbb{R}^n$) can be written uniquely as

$$v = v^{(1)} + \dots + v^{(p)}, \quad v^{(j)} \in V^{(j)} \Rightarrow Av = \lambda_1 v^{(1)} + \dots + \lambda_p v^{(p)}.$$

Observe that each $v^{(j)}$ is a function of v . Indeed, we have $v^{(j)} = \sum_{k=1}^{n_j} \alpha_k^{(j)} v_k^{(j)}$, so

$$v = \sum_{k=1}^{n_1} \alpha_k^{(1)} v_k^{(1)} + \cdots + \sum_{k=1}^{n_p} \alpha_k^{(p)} v_k^{(p)}$$

and thus $\alpha_k^{(j)} = (v_k^{(j)})^* v$, from which

$$v = \sum_{k=1}^{n_1} (v_k^{(1)} (v_k^{(1)})^*) v + \sum_{k=1}^{n_2} (v_k^{(2)} (v_k^{(2)})^*) v + \cdots + \sum_{k=1}^{n_p} (v_k^{(p)} (v_k^{(p)})^*) v.$$

Now, let

$$P_j = \sum_{k=1}^{n_j} v_k^{(j)} (v_k^{(j)})^* . \quad (3.2.4)$$

Then we have $v = (\sum_{j=1}^p P_j) v$ and $Av = (\sum_{j=1}^p \lambda_j P_j) v$. Therefore (since v is arbitrary), these give $\begin{cases} I = \sum_{j=1}^p P_j \\ A = \sum_{j=1}^p \lambda_j P_j \end{cases}$. These two formulas are called *resolution of the identity* and *spectral resolution of A* , respectively.

Note that the resolution of the identity holds no matter what was the matrix A from which we formed the eigenspaces and the P_j 's, whereas in the spectral resolution of A we must use the P_j 's formed from A itself.

The following properties of the P_j 's in (3.2.4) are immediate.

- (1) $P_j P_\ell = 0$, $j \neq \ell$, and $P_j^2 = P_j$. [Obvious, since $P_j = \sum_{k=1}^{n_j} v_k^{(j)} (v_k^{(j)})^*$ and $\{v_k^{(j)}\}_{\substack{k=1:p \\ j=1:n_j}}$ is an orthonormal set.]
- (2) $P_j^* = P_j$. [Again obvious from the form of P_j .]

Therefore, each P_j is *Hermitian (symmetric in the real case) and idempotent*. These two properties characterize an *orthogonal projection*: Each P_j projects (orthogonally) a vector onto the eigenspace $V^{(j)}$ (i.e., onto the subspace spanned by the vectors in $V^{(j)}$). See below.

Example 3.2.23 Suppose $A = \sum_{j=1}^p \lambda_j P_j$ as above.

- (a) Then $A^2 = (\sum_{j=1}^p \lambda_j P_j) (\sum_{j=1}^p \lambda_j P_j) = \sum_{j=1}^p \lambda_j^2 P_j$ and inductively $A^m = \sum_{j=1}^p \lambda_j^m P_j$.
- (b) $q(A) = \sum_{j=1}^p q(\lambda_j) P_j$ for any polynomial q . In fact, the same holds true for any power series in A . For example, if

$$f(A) = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} \Rightarrow e^A = \sum_{j=1}^p e^{\lambda_j} P_j .$$

The foregoing can be generalized.

Definition 3.2.24 (Orthogonal projection) Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$ (or $\mathbb{R}^n$), $1 \leq p \leq n$. A matrix $P \in \mathbb{C}^{n \times n}$ (or $\mathbb{R}^{n \times n}$) such that: $\mathcal{R}(P) = \mathcal{V}$, $P^* = P$, and $P^2 = P$, is called orthogonal projection onto $\mathcal{V}$. The matrix $Q = I - P$ is called complementary projection (or orthogonal complement) to P . We also write $P^\perp$ in lieu of Q . Of course, $PP^\perp = 0$.

The most important consequence of having an orthogonal projection is the possibility to approximate, optimally, from within a certain subspace. Before seeing this, we observe that $\mathbb{C}^n$ can be always thought of as the direct sum of a subspace $\mathcal{V}$ and its orthogonal complement $\mathcal{V}^\perp$: $\mathcal{V}^\perp = \{w \in \mathbb{C}^n : v^*w = 0 \ \forall v \in \mathcal{V}\}$. We write $\mathbb{C}^n = \mathcal{V} \oplus \mathcal{V}^\perp$, and this writing means that any vector in $\mathbb{C}^n$ can be decomposed (uniquely) as the sum of a vector in $v \in \mathcal{V}$ and a vector $v^\perp \in \mathcal{V}^\perp$.

Exercise 3.2.25 Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$, $1 \leq p \leq n$. Show that the decomposition of a vector $x \in \mathbb{C}^n$ as $x = v + v^\perp$, with $v \in \mathcal{V}$ and $v^\perp \in \mathcal{V}^\perp$, is unique.

Remark 3.2.26 Observe that, if P is an orthogonal projection, the subspaces $\mathcal{R}(P)$ and $\mathcal{R}(I - P)$ are orthogonal (which explains the name orthogonal projection). This is easy to appreciate, since if $v \in \mathcal{R}(P)$ and $w \in \mathcal{R}(I - P)$, then $w^*v = w^*(Pv) = w^*P^*v = (Pw)^*v = 0$.

Exercise 3.2.27 Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$, $1 \leq p \leq n$. Let $\{v_1, \dots, v_p\}$ and $\{w_1, \dots, w_p\}$ be two bases for $\mathcal{V}$. Show that

$$\begin{bmatrix} w_1 & \cdots & w_p \end{bmatrix} = \begin{bmatrix} v_1 & \cdots & v_p \end{bmatrix} C, \quad \text{with } C \in \mathbb{C}^{p \times p} \text{ and invertible}.$$

Exercise 3.2.28 Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$, $1 \leq p \leq n$. Let $\{v_1, \dots, v_p\}$ be a basis for $\mathcal{V}$ and form the matrix $V = [v_1, \dots, v_p] \in \mathbb{C}^{n \times p}$. Show that $P = V(V^*V)^{-1}V^*$ is an orthogonal projection onto $\mathcal{V}$ and that this expression is independent of the choice of basis for $\mathcal{V}$. As a consequence, the orthogonal projection onto $\mathcal{V}$ is unique.

Finally, in the next example, we look at the most common use of orthogonal projections.

Example 3.2.29 (On Best Approximation) Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$, $1 \leq p \leq n$, and let $\{v_i\}_{i=1}^p$ be an orthonormal basis for $\mathcal{V}$. Form the matrix $V = [v_1, \dots, v_p]$. Then, we can write $P = VV^*$ for the orthogonal projection onto

$\mathcal{V}$, and its orthogonal complement is $P^\perp = I - P$. We want to solve this frequently arising problem below.

Best approximation out of a subspace

Find $v \in \mathcal{V}$ closest in the 2-norm to a given vector $b \in \mathbb{C}^n$.

That is, we want

$$\min_{v \in \mathcal{V}} \|v - b\|_2^2.$$

Since $v \in \mathcal{V}$, then $v = Pv$ and we can write $b = Pb + (I - P)b = Pb + b^\perp$. So, we have $\|Pb + b^\perp - Pv\|_2^2 = \|P(b - v) + (I - P)b\|_2^2 = (P(b - v) + (I - P)b)^*(P(b - v) + (I - P)b) = \|(I - P)b\|_2^2 + \|P(b - v)\|_2^2 = \|b^\perp\|_2^2 + \|Pb - v\|_2^2$ and the only part we can control in $\mathcal{V}$ is $\|Pb - v\|_2$. As a consequence, the minimizing solution is obtained by choosing $v = Pb$. $\square$

Definition 3.2.30 Let $\mathcal{V}$ be a p -dimensional subspace of $\mathbb{C}^n$, $1 \leq p \leq n$, and let P be the orthogonal projection onto $\mathcal{V}$. Then, the vector $v = Pb$ is called the orthogonal projection of b onto $\mathcal{V}$.

The following consequence of the above is now immediate.

Theorem 3.2.31 In the 2-norm, the orthogonal projection of a vector b onto the subspace $\mathcal{V}$ is the best approximation to b , in the 2-norm, by a vector in $\mathcal{V}$. $\square$

Example 3.2.32 (Overdetermined System) Suppose we have the overdetermined system of equations $Ax = b$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $m > n$, and A is full rank. We seek $x \in \mathbb{R}^n$ such that $\|Ax - b\|_2^2$ is minimized. This is a standard least squares problem, and we know that if A is full rank, then the solution (via a simple calculation) is given by $x = (A^T A)^{-1} A^T b$ (in the notation of Example 3.2.29, it is $v = Ax$). Therefore, $A(A^T A)^{-1} A^T$ must be the orthogonal projection onto the range of A , $\mathcal{R}(A)$. Let

$$P = A(A^T A)^{-1} A^T,$$

and let us verify that this $P \in \mathbb{R}^{m \times m}$ is the orthogonal projection onto $\mathcal{R}(A)$.

- 1) Obviously, $P = P^T$, and $P^2 = A(A^T A)^{-1} A^T A(A^T A)^{-1} A^T = P$.
- 2) Clearly $Px \in \mathcal{R}(A)$, $\forall x \in \mathbb{R}^m$.
- 3) Now, let $b \in \mathcal{R}(A) \Rightarrow b = Ay$, $y \in \mathbb{R}^m$, so we need to show that $\|b - z\|_2$ for $z \in \mathcal{R}(A)$ is minimized by $z = A(A^T A)^{-1} A^T b$. But this is exactly how we derived it: $\|b - z\|_2 = \min_{y \in \mathbb{R}^n} \|b - Ay\|_2 \Rightarrow y = (A^T A)^{-1} A^T b$.

Moreover, we observe that there is a complementary orthogonal projection $P^\perp$ as well, which projects onto the orthogonal complement of $\mathcal{R}(A)$. From the resolution of the identity, $P^\perp = I - P$ and thus $P^\perp = I - A(A^T A)^{-1} A^T$ and clearly $P^\perp P = 0$. Note that $\mathcal{R}(A)^\perp = \mathcal{N}(A^T)$ (the null space of A^T). This implies that (see Exercise 3.2.33)

“ $Ax = b$ has a solution if and only if $b^T w = 0$, for all $w \in \mathcal{R}(A)^\perp$, that is for all w : $A^T w = 0$.”

Finally, we remark that it is not desirable to form the projection P as $A(A^T A)^{-1} A^T$ and a much sounder approach is the following. We want to form the orthogonal projection onto $\mathcal{R}(A)$, the column space of A . So, we can take (say, using the Gram-Schmidt process) an orthonormal basis for $\mathcal{R}(A)$ by taking the QR factorization of A itself (see Theorem 2.1.8): $A = QR$, and the columns of Q provide an orthonormal basis for $\mathcal{R}(A)$. Thus, see Exercise 3.2.27 we can form the orthogonal projection onto $\mathcal{R}(A)$ by taking $P = QQ^T$ (and of course $P^\perp = I - QQ^T$). $\square$

Exercise 3.2.33 Show that $\mathcal{R}(A)^\perp = \mathcal{N}(A^T)$ (the null space of A^T , also called left null space of A) and that this implies that $Ax = b$ has a solution if and only if $b^T w = 0$, for all $w \in \mathcal{R}(A)^\perp$.

As it turns out, every orthogonal projection has a very simple canonical form. Below we find this form working in $\mathbb{R}$, but the same result holds in $\mathbb{C}$. To begin with, we have the following result.

Exercise 3.2.34 Let P be the orthogonal projection onto a nontrivial subspace $\mathcal{V}$ of $\mathbb{R}^n$. Then: $\|P\|_2 = 1$ and $\lambda_{\max}(P) = 1$.

Solution. Take x and write it as $x = x_P + x_{P^\perp} = Px + (I - P)x \Rightarrow \|Px\|_2 = \|Px_P\|_2 = \|x_P\|_2$. Therefore,

$$\max_{x \in \mathbb{R}^n, \|x\|=1} \|Px\|_2^2 = \max_{x \in \mathcal{R}(P), \|x\|=1} \|Px\|_2^2 = \max_{x \in \mathcal{R}(P), \|x\|=1} x^T P P x = \max_{\|x_P\|=1} x_P^T x_P = 1,$$

and so $\|P\|_2 = 1$; further, P has largest eigenvalue equal to 1, since $\|P\|_2^2 = \lambda_{\max}(P^T P) = \lambda_{\max}(P^2) = \lambda_{\max}(P)$. $\square$

Note that, in Exercise 3.2.34 we have used that for any $x \in \mathcal{R}(P)$ we have $Px = x$ and for any $x \in \mathcal{R}(I - P)$ we have $Px = 0$. This means that P has as many eigenvalues equal to 1 as $\text{rank}(P)$, the rest being all 0's.

The following corollary is now an immediate consequence of symmetry, spectral theorem for symmetric matrices, and rank of an orthogonal projection onto a p -dimensional subspace.

Corollary 3.2.35 *After an orthogonal change of coordinates, an orthogonal projection onto a p -dimensional subspace can be written as $P = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}$. $\square$*

Example 3.2.36 Let $A = \frac{1}{n} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 1 \end{bmatrix}$. Clearly A is an orthogonal projection ($A^T = A$, $A^2 = A$) and has rank 1 and therefore A has one eigenvalue equal to 1 and $(n-1)$ eigenvalues equal to 0. Given that all columns of A span the same line, obviously A projects onto this very line.

Indeed, if $Q : Q^T A Q = \Lambda$, with $\Lambda = \begin{bmatrix} 1 & & & \\ & 0 & & \\ & & \ddots & \\ & & & 0 \end{bmatrix}$, then $A = q_1 q_1^T$ (and $q_1 = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$).

$\square$

Exercise 3.2.37 (Non orthogonal projections) *We could have defined as projection onto a p -dimensional subspace $\mathcal{V}$ of $\mathbb{C}^n$ any matrix $B \in \mathbb{C}^{n \times n}$ of rank p such that $B^2 = B$ (idempotency), without requiring symmetry.*

- (a) *Show that a matrix B (not necessarily symmetric) is a projection if and only if it is similar to $\begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}$.*
- (b) *If V is such that $V^{-1} B V = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}$, then show that B projects onto the subspace $\mathcal{V}$ spanned by the first p columns of V . This shows that $Bx = x$ for any $x \in \mathcal{V}$.*
- (c) *In $\mathbb{R}^3$, build the orthogonal projection P onto the subspace given by the plane $2x_1 - 3x_2 + x_3 = 0$. Also, build another nonorthogonal projection B onto this same subspace.*
- (d) *In $\mathbb{R}^2$, give an example of a (nonorthogonal) projection B onto one of the coordinate axes, with B of arbitrarily large 2-norm.*
- (e) *Let $B \in \mathbb{C}^{n \times n}$ be a projection, in that $B^2 = B$. Show that B is an orthogonal projection, that is $B^* = B$ if and only if $\|Bv\|_2 \leq \|v\|_2$ for every $v \in \mathbb{C}^n$.*

Exercise 3.2.38 Let $A = A^*$ have eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > \lambda_{p+1} \geq \cdots \geq \lambda_n$, and let V be an orthonormal matrix whose columns span the same subspace as the p eigenvectors associated to $\{\lambda_1, \dots, \lambda_p\}$ [dominant invariant subspace]. Let P be the orthogonal projection onto $\mathcal{R}(V)$. Suppose we want to compute $e^A v$, with $v \in \mathbb{C}^n$, $\|v\|_2 = 1$, and where $Pv \neq 0$. Find a formula for the relative error $\|e^A v - e^A P v\|_2 / \|e^A v\|_2$ (recall Example 3.2.23-(b)), involving the eigenvalues of A .

3.2.2 Positive definiteness and congruence

An important class of Hermitian (symmetric) matrices are those matrices which have all positive eigenvalues.

Definition 3.2.39 $A \in \mathbb{C}^{n \times n}$ (respectively, $A \in \mathbb{R}^{n \times n}$), $A^* = A$ (respectively, $A^T = A$), is called positive definite if $\langle x, Ax \rangle > 0$, $\forall x \neq 0$. It is called nonnegative definite, or positive semi-definite, if $\langle x, Ax \rangle \geq 0$, $\forall x$. It is called negative definite, if $\langle x, Ax \rangle < 0$, $\forall x \neq 0$, and nonpositive definite, or negative semi-definite, if $\langle x, Ax \rangle \leq 0$, $\forall x$. In these four cases, we also write $A > 0$, $A \geq 0$, $A < 0$, $A \leq 0$, respectively.

- Hermitian (symmetric) matrices that are not definite in any of the four stated meanings, are called *indefinite*. That is, indefinite means that A is Hermitian, but there exist $x, y \in \mathbb{C}^n$, such that $\langle x, Ax \rangle > 0$ and $\langle y, Ay \rangle < 0$.

Remark 3.2.40 Be aware that we only defined positive definite matrices which are Hermitian (symmetric). One may possibly define positive definite non-Hermitian (non-symmetric) matrices as those matrices A for which $\langle x, Ax \rangle > 0$, $\forall x \neq 0$; for example, the real matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ would be positive definite. However, many of the most useful properties of positive definite matrices, like having positive eigenvalues (see Theorem 3.2.45), would not generally hold if we remove the condition $A^* = A$.

Exercises 3.2.41 (Properties of positive definite matrices)

- (1) (a) Show that if A, B are positive definite $\Rightarrow A + B$ is too. (b) Show that if A is positive definite, and $\alpha \in \mathbb{R}$, $\alpha > 0$, then αA is also positive definite. [Note: These two properties say that the set of positive definite matrices is a **cone**.]
- (2) Show that if A is definite (positive, negative, nonpositive or nonnegative), and S is any invertible matrix, then $S^* A S$ is definite of the same type as A . [We will

encounter again this type of transformation of A , it is a “congruence” relation.] In particular, you have shown that positive definiteness is preserved by unitary similarity.

- (3) Show that A is positive definite $\Leftrightarrow$ all its eigenvalues are positive. Similarly, A is negative definite $\Leftrightarrow$ all its eigenvalues are negative.

As consequences of this fact, for positive definite matrices we have:

- a) $\langle x, Ax \rangle \geq \lambda_{\min}(A) \|x\|^2$, $\forall x$, where $\lambda_{\min}(A)$ is the smallest eigenvalue of A .
 b) A is invertible and A^{-1} is also positive definite.

Lemma 3.2.42 (Open cone) Consider the set $\mathcal{P} = \{A \in \mathbb{C}^{n \times n} : A^* = A \text{ and } A \text{ is positive definite}\}$. Then $\mathcal{P}$ is an open set in the set of Hermitian matrices. Similarly for real symmetric positive definite matrices.

Pf. Let $A = A^*$ be positive definite with ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$. Let $B = B^*$ be such that $\|A - B\|_2 < \lambda_n$. We claim that B is positive definite. (Notice that, then, it will follow that $\mathcal{P}$ is open).

Let $M = A - B$. We know that for any $x \neq 0$ (the second inequality is Exercise 3.2.41-(3))

$$\begin{cases} \|Mx\|_2 < \lambda_n \|x\|_2 \\ \langle x, Ax \rangle \geq \lambda_n \|x\|_2^2 \end{cases}.$$

So, for any $x \neq 0$, we have:

$$0 \leq |\langle x, Mx \rangle| \leq \|x\|_2 \cdot \|Mx\|_2 < \lambda_n \|x\|_2^2,$$

and also

$$\begin{aligned} \langle x, Bx \rangle &= \langle x, (A - M)x \rangle = \langle x, Ax \rangle - \langle x, Mx \rangle \geq \lambda_n \|x\|_2^2 - \langle x, Mx \rangle \\ &\Rightarrow \langle x, Bx \rangle > \lambda_n \|x\|_2^2 - \lambda_n \|x\|_2^2 = 0. \end{aligned}$$

Therefore, $\langle x, Bx \rangle > 0$, $\forall x \neq 0$ and so B is positive definite. $\square$

Lemma 3.2.43 (Boundary of Cone) The boundary elements of the set of positive-definite matrices are the nonnegative definite matrices, which are not positive definite.

Pf. We look at matrices which are limits of sequences $\{A_k\}$ of positive definite matrices:

$$B = \lim_{k \rightarrow \infty} A_k, \quad A_k = A_k^* \text{ positive definite.}$$

Now, $\forall x \neq 0 : \langle x, A_k x \rangle > 0 \therefore \lim_{k \rightarrow \infty} \langle x, A_k x \rangle \geq 0 \therefore \langle x, Bx \rangle \geq 0, \forall x \in \mathbb{C}^n \therefore B$ is nonnegative definite. But if B is positive definite, then by Lemma 3.2.42 it could not be a boundary element. Finally, if B is a nonnegative definite matrix, then we claim that it is on the boundary of the cone. To verify this claim, it is enough to take the sequence of positive definite matrices given by $A_k = B + \frac{1}{k}I$. $\square$

Lemmata 3.2.42 and 3.2.43 (which holds also for positive definite matrices in $\mathbb{R}^{n \times n}$) give us the geometrical description of the set of positive definite matrices, to which we simply add that the vertex of the cone is the origin (the 0-matrix).

Let us now verify some important properties of positive definite matrices. The next results are given for positive definite Hermitian matrices, but everything remains true for positive definite matrices in $\mathbb{R}^{n \times n}$, with obvious minimal changes; e.g., A^T instead of A^* , etc..

Lemma 3.2.44 (Positive definite square root) *If $A \in \mathbb{C}^{n \times n}$ is positive definite, then there exists a unique matrix $B \in \mathbb{C}^{n \times n}$, positive definite, such that $B^2 = A$.*

Pf. $(\exists)$ Let U be such that $U^*AU = D$ is diagonal and positive definite: $D = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}, d_i > 0$. Then, define the (diagonal and positive definite) matrix $D^{1/2} = \begin{pmatrix} d_1^{1/2} & & 0 \\ & \ddots & \\ 0 & & d_n^{1/2} \end{pmatrix}$, so that $(D^{1/2})^2 = D$. Now take $B = UD^{1/2}U^*$, which is clearly positive definite and satisfies $B^2 = A$. Note that since $B^2 = A$, and B is invertible, then $B = B^{-1}A$, and thus $AB = B^2B^{-1}A = BA$, and A and B commute.

$(!)$ First of all, we observe that the matrix B above is unchanged under column permutations of U . In other words, if P is a permutation matrix associated to the permutation $\sigma(i), i = 1, \dots, n$, then $P^T U^* A U P = P^T D P$, which is nothing but the diagonal matrix $\widehat{D}$ of entries $d_{\sigma(i)}, i = 1, \dots, n$. So, the unitary matrix UP has diagonalized A , and moreover $B = UD^{1/2}U^* = (UP)\widehat{D}^{1/2}(UP)^*$. Therefore, if $\mu_1, \dots, \mu_p$, are the distinct eigenvalues of A , of algebraic multiplicities $n_1, \dots, n_p$, we can assume without loss of generality that the unitary matrix U gave the decomposition

$$U^*AU =: \widehat{D} = \begin{bmatrix} \mu_1 I_{n_1} & & \\ & \ddots & \\ & & \mu_p I_{n_p} \end{bmatrix}, \text{ and } B = U \widehat{D}^{1/2} U^*.$$

Next, suppose that there exists $C = C^*$, positive definite, such that $C^2 = A$, and thus we have both $C^3 = CA$ and $C^3 = AC$ and so A and C commute. Therefore, A and C are simultaneously diagonalizable, and being Hermitian, they are

simultaneously diagonalizable by a unitary transformation (see Exercise 3.2.10). Therefore, there exists Z unitary such that $Z^*AZ = \widehat{D}$, and $Z^*CZ = D_C$, from which $D_c = \widehat{D}^{1/2}$. Therefore, from Exercise 3.2.9 (part (a)), we must have $Z =$

$$U \begin{bmatrix} V_1 & & \\ & \ddots & \\ & & V_p \end{bmatrix}, \quad V_j^* V_j = I_{n_j}, \text{ from which (and the form of } \widehat{D}) \text{ one immediately gets}$$

that $C = ZD_cZ^* = Z\widehat{D}^{1/2}Z^* = U\widehat{D}^{1/2}U^* = B$. $\square$

- There are several other useful characterizations of positive definite matrices. We recall a few below.

Theorem 3.2.45 *Let $A = A^*$. The following statements are equivalent.*

- (0) A is positive definite.
- (1) $\exists C$ nonsingular, such that $A = CC^*$.
- (2) $\exists L$, lower triangular with positive diagonal such that $A = LL^*$.
- (3) All eigenvalues of A are positive.
- (4) All the submatrices $A(1:k, 1:k)$, $k = 1, 2, \dots, n$, have positive determinant. The matrices $A(1:k, 1:k)$, $k = 1, 2, \dots, n$, are called leading principal minors of A .

Statement (1) in Theorem 3.2.45 is another instance of a congruence relation (see below). Statement (2) gives the *Choleski factorization* of A .

Exercises 3.2.46

- (1) Show that if A is positive definite $\Rightarrow$ it has a unique positive definite k -th root, for any $k \in \mathbb{Z}$, $k \neq 0$.
- (2) Prove Theorem 3.2.45, realizing that Statement (3) was already the content of Exercise 3.2.41-(3).
- (3) Show that if $A = A^*$ is nonnegative definite, then it has a unique nonnegative definite square root.
- (4) [Signed Choleski] Show that $A = A^*$ is positive definite if and only if there exists L , lower triangular and nonsingular, such that $A = LL^*$.
- (5) Let $A \in \mathbb{C}^{n \times n}$ be positive definite. How many Hermitian square roots of A are there? How many Hermitian cubic roots?
- (6) Let $A \in \mathbb{C}^{2 \times 2}$ be nonnegative definite. How many Hermitian square roots of A are there?
- (7) Given some $A = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \gamma \end{pmatrix} \in \mathbb{C}^{2 \times 2}$ and positive definite. Can A have any non-Hermitian square root (that is, can there be a matrix B , non Hermitian, such that $B^2 = A$)?

(8) Here, we use the notation from Example 3.2.3, and consider the case where the Hessian is definite. The indefinite case is introduced in Example 5.2.57.

(a) Show that if x_0 is a critical point of the scalar function f , and the Hessian evaluated at x_0 is positive definite, then x_0 is a strict minimum of f . Similarly, if H is negative definite, then x_0 is a strict maximum.

(b) Next, suppose that the function f is a real valued quadratic function, that is $f(x) = \frac{1}{2}x^T Hx - x^T b + c$. Show that if the Hessian H is nonnegative definite, and b is in the column space of H , then there is a subspace of critical points all giving the same minimal value of f . Characterize the dimension of such subspace. (Similarly, if H is nonpositive definite there is a subspace of critical points giving the same maximal value of f .)

Exercise 3.2.47 Nonnegative definite matrices have similar characterizations to positive definite matrices. Show Theorem 3.2.48 below.

Theorem 3.2.48 Let $A = A^*$. The following statements are equivalent:

- (0) A is nonnegative definite.
- (1) $\exists C$ such that $A = CC^*$.
- (2) All eigenvalues of A are non negative.
- (3) All leading principal minors of A have non negative determinant.

- Another useful characterization of positive definiteness is in terms of the Gram-matrix.

Definition 3.2.49 Given vectors $v_1, \dots, v_p$ in $\mathbb{C}^n$, the matrix $G \in \mathbb{C}^{p \times p}$ of entries $G_{ij} = \langle v_j, v_i \rangle$ is called Gram matrix (or Gramian) of the given vectors. If we let $V = [v_1 \ v_2 \ \dots \ v_p] \in \mathbb{C}^{n \times p}$, then $G = V^*V$.

Observation 3.2.50 Obviously, the Gramian is a Hermitian matrix.

Theorem 3.2.51 For a Gram matrix G , the following hold.

- 1) G is nonnegative definite.
- 2) The vectors $\{v_1, \dots, v_p\}$ are linearly independent $\iff$ their Gramian G is positive definite.
- 3) Every nonnegative definite matrix $A \in \mathbb{C}^{n \times n}$ is the Gram matrix of n vectors in $\mathbb{C}^n$.

Pf. For any vector x , we have

$$\begin{aligned}\langle x, Gx \rangle &= x^* G^* x = x^* G x = \sum_{i,j=1}^p G_{ij} \bar{x}_i x_j = \sum_{i,j=1}^p \langle v_j, v_i \rangle \bar{x}_i x_j \\ &= \sum_{i,j=1}^p \langle x_j v_j, x_i v_i \rangle = \left\langle \sum_{j=1}^p x_j v_j, \sum_{i=1}^p x_i v_i \right\rangle = \left\| \sum_{i=1}^p x_i v_i \right\|^2\end{aligned}$$

and so 1) and 2) follow at once.

To get 3), let B be such that $B^2 = A$, $B = B^*$ (see Exercise 3.2.46-(3)). Then $A = B^* B \Rightarrow A_{ij} = \langle b_j, b_i \rangle$, $i, j = 1, \dots, n$. $\square$

Exercises 3.2.52

- (1) Suppose we have two set of vectors $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_n\}$ in $\mathbb{C}^n$ giving the same Gramian. What is the relation between the matrices $V = [v_1, \dots, v_n]$ and $W = [w_1, \dots, w_n]$? [Hint: You may want to first assume that the matrices are invertible. Then, it may be simpler to do this exercise after Section 5.1.2.]
- (2) How does the Gramian of a set of mutually orthogonal vectors look like? Of an orthonormal set?

Example 3.2.53 [Hilbert matrix] Recall that $C([0, 1], \mathbb{R})$ is the space of continuous function from $[0, 1]$ to the real numbers. Now, consider $C([0, 1], \mathbb{R})$ with the usual L_2 inner product $(f, g) = \int_0^1 f(t)g(t)dt$, and norm $\|f\|^2 = \int_0^1 f^2(t)dt$. Take the functions $f_j = t^{j-1}$, $j = 1, 2, \dots$, clearly linearly independent and nonnegative in $[0, 1]$. Consider the Gram matrix of entries $G_{ij} = (f_j, f_i)$, $i, j = 1, \dots, n$, so that $(f_j, f_i) = G_{ij} = \frac{1}{i+j-1}$. In this case, the Gramian is the famous Hilbert matrix, which is positive definite (see Exercise 3.2.54-(a)). The Hilbert matrix is a standard example of a severely ill-conditioned matrix for large n . The condition number of G , $\|G\|\|G^{-1}\|$, gets large for n large.

Exercise 3.2.54

- (a) Show that the Hilbert matrix is positive definite.
- (b) (Harder) Find and understand a proof deriving the explicit expression for the inverse of the Hilbert matrix. In particular, this will show that the inverse has only integer entries. [Hint: If you want to challenge yourself and derive a proof on your own, it is best to change basis for the polynomials, from the monomial basis to the Legendre polynomials.]

Exercise 3.2.55 [Hankel matrix] *Given a positive function f , $f(t) > 0$ for $t \in [0, 1]$, consider the algebraic moments*

$$a_k = \int_0^1 t^k f(t) dt, \quad k = 0, 1, 2, \dots$$

To these, we associate the quadratic form

$$\sum_{j,k=0}^n a_{j+k} x_j x_k = \sum_{j,k=0}^n \int_0^1 t^{j+k} x_j x_k f(t) dt,$$

and the matrix $A \in \mathbb{R}^{n+1, n+1}$ defined by $(A)_{ij} = a_{i+j}$. Notice that the entries of A are functions only of $i + j$. Every time this property holds for a matrix A , A is called a Hankel matrix. (The Hilbert matrix is a special Hankel matrix). Show that A is positive definite.

Exercise 3.2.56 [Toeplitz matrix] *Consider the trigonometric moments:*

$$a_k = \int_0^1 e^{ikt} f(t) dt, \quad k = 0, \pm 1, \pm 2, \dots,$$

and the associated quadratic form: $\sum_{j,k=0}^n a_{j-k} z_j \bar{z}_k$. Form the matrix A and verify that its entries only depend on $i - j$. These are called Toeplitz matrices. Show that if $f > 0$, then A is positive definite.

- Statement (1) of Theorem 3.2.45 told us that $A = A^*$ is positive definite $\Leftrightarrow A = CC^*$ for an invertible matrix C . By looking at this as $C^{-1}AC^{-*} = I$, then we say that A is *congruent* to the identity.

Definition 3.2.57 (Congruence) *Given $A = A^*$ and S invertible, then $B = SAS^*$ is called congruent to A . Similarly, for real matrices: $B = SAS^T$.*

Observe that congruence is an equivalence relation. Of course, congruence is not the same as similarity (unless S is unitary), but it still allows for some simplifications in many problems.

Remark 3.2.58 *It is insightful to think of similarity as a simplification of the “dynamics” of a problem. For example, consider the discrete dynamical system $y^{(k+1)} = Ay^{(k)}$. The change of variable $v = V^{-1}y$ gives $v^{(k+1)} = (V^{-1}AV)v^{(k)}$ and V should be chosen so that $V^{-1}AV$ is simpler (say, in Jordan form). In some contexts, congruence arises naturally when we change the “internal” variables of a system, as the next example clarifies.*

Example 3.2.59 Let f be a twice continuously differentiable function in some domain D of $\mathbb{R}^n$, and consider the 2nd order linear differential operator L :

$$Lf(x) = \sum_{i,j=1}^n a_{ij}(x) \frac{\partial^2 f(x)}{\partial x_i \partial x_j}.$$

To this operator, we associate the matrix $A = (a_{ij})$, where the entries a_{ij} are functions of $x \in D$. Observe that A may fail to be symmetric, but since

$$\begin{aligned} \sum_{i,j=1}^n a_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} &= \sum_{i,j=1}^n \left[\frac{1}{2} a_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} + \frac{1}{2} a_{ji}(x) \frac{\partial^2 f}{\partial x_j \partial x_i} \right] \\ &= \sum_{i,j=1}^n \frac{1}{2} (a_{ij}(x) + a_{ji}(x)) \frac{\partial^2 f}{\partial x_i \partial x_j}, \end{aligned}$$

then L is impacted only by the symmetric part of A and so we may as well assume that A is symmetric.

Now, suppose that we make a nonsingular change of variables from x to $s = (s_i)_{i=1}^n$, with $s_i = s_i(x_1, \dots, x_n)$. Nonsingularity implies that the Jacobian matrix $S = \left(\frac{\partial s_i(x)}{\partial x_j} \right)_{i,j=1}^n$ is nonsingular (and also that the change of variable is invertible: $x = x(s)$, at least locally). Under this change of variables (**Exercise:** Verify (lengthy, but easy)), we get

$$Lf = \sum_{i,j=1}^n \underbrace{\left(\sum_{p,q=1}^n \frac{\partial s_i}{\partial x_p} a_{pq} \frac{\partial s_j}{\partial x_q} \right)}_{b_{ij}(s)} \frac{\partial^2 f}{\partial s_i \partial s_j}.$$

Thus, if we let $B = (b_{ij})_{i,j=1}^n$, then $B = SAS^T$ and B is congruent to A . We expect that physical laws described by the original differential equation will not be impacted by this change of coordinates. $\square$

Motivated by the above, we now address the following questions:

- (1) What are the invariants of congruence transformations?
- (2) What kind of simplifications can we achieve by congruence?

The answers turn out to be remarkably simple.

Definition 3.2.60 (Inertia) Let $A = A^* \in \mathbb{C}^{n \times n}$ (or $A = A^T \in \mathbb{R}^{n \times n}$). We define inertia of A , and write it $i(A)$, to be the triplet $i(A) = (n_+(A), n_-(A), n_0(A))$ where
 $n_+(A) = \#$ positive eigenvalues of A ,
 $n_-(A) = \#$ negative eigenvalues of A ,

$n_0(A) = \#$ zero eigenvalues of A .

Further, we define inertia matrix of A the matrix $I(A) = \begin{bmatrix} I_{n_+(A)} & & \\ & -I_{n_-(A)} & \\ & & 0_{n_0(A)} \end{bmatrix}$.

Remark 3.2.61 The pair $(n_+(A), n_-(A))$ is often called signature of A . But, be aware that there is no consensus on this terminology.

Theorem 3.2.62 (Inertia Matrix) Let $A = A^* \in \mathbb{C}^{n \times n}$ (or $A = A^T \in \mathbb{R}^{n \times n}$), and let $i(A) = (n_+(A), n_-(A), n_0(A))$ be its inertia. Then, A is congruent to its own inertia matrix $I(A)$.

Pf. Let U be such that $U^*AU = \Lambda$ where Λ is the diagonal matrix of eigenvalues, and where in Λ we first put all positive eigenvalues then all the negative ones, then all the 0 ones. Accordingly, define the diagonal and positive definite matrix D :

$$D = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_{n_+}}, \sqrt{-\lambda_{n_++1}}, \dots, \sqrt{-\lambda_{n_++n_-}}, 1, \dots, 1) \\ \Rightarrow \Lambda = DI(A)D \Rightarrow A = (UD)I(A)(DU^*) .$$

Now, let $S = UD$. □

Theorem 3.2.63 (Sylvester law of inertia) Two real symmetric (complex Hermitian) matrices are congruent iff they have same inertia, and therefore the same inertia matrix.

Pf. Consider the real case.

Because of Theorem 3.2.62, we will let S_A and S_B be such that $S_AI(A)S_A^T = A$ and $S_BI(B)S_B^T = B$.

($\Leftarrow$) If $i(B) = i(A) \Rightarrow I(B) = I(A)$, and therefore $B = S_BI(B)S_B^T = S_BI(A)S_B^T = S_BS_A^{-1}S_AI(A)S_A^T(S_A^T)^{-1}S_B^T \Rightarrow B = (S_BS_A^{-1})A(S_BS_A^{-1})^T$
 $\therefore A$ and B are congruent.

($\Rightarrow$) Let S invertible be such that $A = SBS^T$. Obviously, $\text{rank}(A) = \text{rank}(B)$ and (since A and B are symmetric) $\text{rank}(A) = n - n_0(A) = n - n_0(B) = \text{rank}(B)$, so that $n_+(A) + n_-(A) = n_+(B) + n_-(B)$. Now we show that $n_+(A) = n_+(B)$. Note:

$$I(A) = \begin{bmatrix} I_{n_+(A)} & & \\ & -I_{n_-(A)} & \\ & & 0_{n_0} \end{bmatrix}, \quad I(B) = \begin{bmatrix} I_{n_+(B)} & & \\ & -I_{n_-(B)} & \\ & & 0_{n_0} \end{bmatrix},$$

and since $A = S_A I(A) S_A^T$, $B = S_B I(B) S_B^T$, then we have $I(B) = T^T I(A) T$, with $T = S_B^T S^T S_A^{-T}$ invertible, and therefore $I(A)$ and $I(B)$ are congruent.

So, we want to show that two congruent inertia matrices have the same inertia.

Suppose not. Then, without loss generality $n_+(A) < n_+(B)$. Consider the subspace $V = \left\{ v \in \mathbb{R}^n : v = \begin{bmatrix} 0 \\ v_- \\ 0 \end{bmatrix} \begin{matrix} \} n_+(A) \\ \} n_-(A) \\ \} n_0 \end{matrix} \right\}$ and notice that $\dim(V) = n_-(A)$ and

$\langle v, I(A)v \rangle < 0, \forall v \neq 0, v \in V$. Consider also the subspace $W = \left\{ w \in \mathbb{R}^n : w = \begin{bmatrix} w_+ \\ 0 \\ w_0 \end{bmatrix} \begin{matrix} \} n_+(B) \\ \} n_-(B) \\ \} n_0 \end{matrix} \right\}$

and notice that $\dim(W) = n_+(B) + n_0 = \dim(TW)$, since T is invertible. Moreover, $0 \leq \langle w, I(B)w \rangle = \langle w, T^T I(A) T w \rangle = \langle Tw, I(A)Tw \rangle, \forall w \in W$.

Now: $\dim(V) + \dim(TW) = n_-(A) + n_0 + n_+(B) > n \Rightarrow \exists x \neq 0, x \in V \cap TW$. But then for this x we would have

$$\begin{aligned} \langle x, I(A)x \rangle &< 0, & (x \in V) \\ \langle x, I(A)x \rangle &\geq 0, & (x \in TW) \end{aligned}$$

which is a contradiction and therefore $n_+(A) = n_+(B)$.

The proof in the complex case is similar, since $\langle x, I(A)x \rangle = \langle I(A)x, x \rangle$. $\square$

Corollary 3.2.64 *Under congruence, we have a partitioning of the set of Hermitian (symmetric) matrices into equivalence classes, each having the same inertia, and hence the same inertia matrix.* $\square$

Exercise 3.2.65 *Given the set $H = \{A \in \mathbb{C}^{n \times n}, A^* = A\}$. How many equivalence classes are there for H under congruence?*

Remark 3.2.66 *An interesting question, also with physical motivation, is to decide if/when two matrices can be simultaneously diagonalized by congruence. (Note: not necessarily to the same inertia matrix.)*

A typical result reads as follows (for a proof, and similar statements, see [5]): “Let $A = A^*$, $B = B^*$ and A be nonsingular. Form $C = A^{-1}B$. Then, there exists invertible S such that SAS^* and SBS^* are both diagonal $\Leftrightarrow C$ is diagonalizable and has real eigenvalues.”

3.2.3 More inner products and projections

One natural way in which congruence transformations arise is when we work with an inner product (hence, a norm) induced by a positive definite matrix different from the identity. The starting point is to realize that, whereas the identity matrix defines the Euclidean metric, any other positive definite matrix can be used to define a metric. In fact, any positive definite matrix can be used to define an inner product, and hence a norm and a metric. (Cfr. with Riemannian metric (inner product) in differential geometry; see [22]).

Formally, given a vector space $\mathcal{V}$ over a field $\mathbb{F}$ (either $\mathbb{C}$ or $\mathbb{R}$), an inner-product is a function

$$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{F}$$

satisfying:

$$\langle x, y \rangle = \overline{\langle y, x \rangle}, \quad \langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle, \quad \text{and} \quad \langle x, x \rangle > 0, \quad \text{if } x \neq 0.$$

Definition 3.2.67 (Positive definite inner product) Let $G = G^* \in \mathbb{C}^{n \times n}$ (or $G = G^T \in \mathbb{R}^{n \times n}$) be positive definite. The quantity

$$\langle x, y \rangle_G = y^* G x$$

is the inner product associated to G or just G -inner product.

The G -norm of a vector $x \in \mathbb{C}^n$ is given by $\|x\|_G = (x^* G x)^{1/2}$.

Observe that –letting $G^{1/2}$ to be the unique positive definite square root of G – the G -inner product can be appreciated to be a weighted standard inner product: $\langle x, y \rangle_G = \langle G^{1/2} x, G^{1/2} y \rangle_2$. Since $G^{1/2}$ is diagonalizable with a unitary transformation, and the diagonal is positive, in the appropriate system of coordinates, we are just assigning positive (and typically different) weights to the different coordinates. For this reason, G -norms are also called *ellipsoidal norms*.

Example 3.2.68 Consider $\mathbb{R}^3$, and let $G = \begin{bmatrix} 2.3104 & 1.2672 & -0.3840 \\ 1.2672 & 3.0496 & 0.2880 \\ -0.3840 & 0.2880 & 1.6499 \end{bmatrix}$. In figure

3.1 are visualizations of the sets $\{x \in \mathbb{R}^3 : \|x\|_2^2 = 1\}$ (the standard unit sphere) and $\{x \in \mathbb{R}^3 : \|x\|_G^2 = 1\}$. The latter is an ellipsoid, though both are unit spheres in their respective metric. In the coordinate system of its principal axes, the ellipsoid is simply $x_1^2 + 2x_2^2 + 4x_3^2 = 1$.

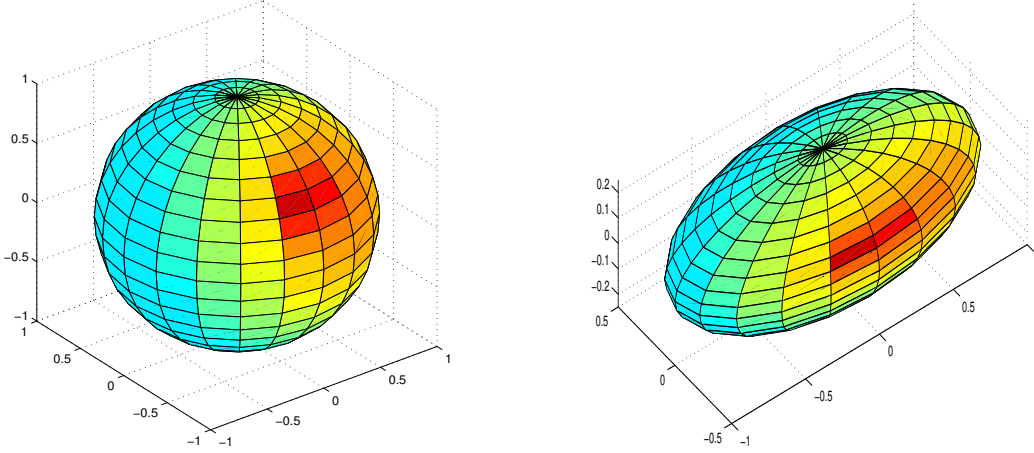


Figure 3.1: The unit sphere in the standard metric and in that induced by G of Example 3.2.68.

It is a worthwhile exercise to verify that the constructions we carried out with respect to the standard inner product can be performed with respect to a G -inner product. For example, the Cauchy-Schwartz inequality now generalizes to:

$$|\langle x, y \rangle_G| \leq \|x\|_G \|y\|_G,$$

and as a consequence we can also talk of angles between vectors from

$$\cos \theta = \frac{\operatorname{Re}(\langle x, y \rangle_G)}{\|x\|_G \|y\|_G},$$

and we have the definition of G -orthogonality:

$$x \text{ and } y \text{ are } G\text{-orthogonal} \iff \langle x, y \rangle_G = 0.$$

With this, we can define vectors to be G -orthonormal, and extend the Gram-Schmidt process with respect to the G -inner product. Likewise, a matrix $U \in \mathbb{C}^{n \times n}$ will be called G -unitary if $U^* G U = I$. Schur's theorem and the like also have a natural extension to this new setting. As illustration, in the next result we look at the G -unitary version of the spectral theorem for Hermitian matrices. You will recognize in it a special congruence transformation.

Theorem 3.2.69 (G -unitary spectral theorem) *Let $A \in \mathbb{C}^{n \times n}$, $A^* = A$, and let $G \in \mathbb{C}^{n \times n}$ be positive definite. Then, there exist $U \in \mathbb{C}^{n \times n}$, G -unitary, such that $U^* A U = D$ and D is diagonal and real. The entries on the diagonal of D can appear in any desired order.*

Pf. Let $G^{1/2}$ be the unique positive definite square root of G , and define $B = G^{-1/2}AG^{-1/2}$, which is Hermitian. Let V , unitary, be such that $V^*BV = D$, the real diagonal matrix of eigenvalues of B , with the eigenvalues ordered in the desired way. Now let $U = G^{-1/2}V$, so that $U^*GU = I$ and $U^*AU = D$. $\square$

Remark 3.2.70 Notice that we do not get the eigenvalues of A from a result like Theorem 3.2.69, since $U^* \neq U^{-1}$ (unless $G = I$). All we can say is that the matrix D of Theorem 3.2.69 has the same inertia as A (in fact, it is trivial to get $I(A)$ from D), and that it is similar to $G^{-1/2}AG^{-1/2}$.

Remark 3.2.71 Observe that the matrix U in Theorem 3.2.69 is the product of a positive definite matrix and a unitary one. This is a polar factorization, see Section 5.1.

Exercise 3.2.72 Consider the notation of Theorem 3.2.69, and recall Exercises 3.1.7 and 3.2.9.

- (1) Let G be positive definite, and let the eigenvalues of D be distinct, appearing along D 's diagonal as $d_1 > d_2 > \dots > d_n$. What freedom do you have in choosing U ?
- (2) Now assume that the eigenvalues of D are not distinct, but that equal eigenvalues of D occur consecutively along D 's diagonal. What freedom do you have in choosing U ?

Once we have Theorem 3.2.69, we can define projections as well, though it is not obvious how to do it.

In the notation of Theorem 3.2.69, let $U^*AU = D$ be a G -unitary diagonalization of a Hermitian matrix $A \in \mathbb{C}^{n \times n}$. Let $d_1, \dots, d_p$, be the distinct entries on the diagonal of D , of multiplicity $n_1, \dots, n_p$, respectively ($n_1 + \dots + n_p = n$), and let $u_k^{(j)}$, $j = 1, \dots, p$, $k = 1, \dots, n_j$, be the columns of U corresponding to the values d_j in D , $j = 1, \dots, p$. So, we have the subspace decomposition of $\mathbb{C}^n$:

$$\mathbb{C}^n = U^{(1)} \oplus U^{(2)} \oplus \dots \oplus U^{(p)} , \quad (3.2.5)$$

where $U^{(j)} = \text{span}(u_1^{(j)}, \dots, u_{n_j}^{(j)})$, $j = 1, \dots, p$.

Now, we claim that the following are orthogonal projections in the usual sense:

$$P_j = \sum_{k=1}^{n_j} G^{1/2} u_k^{(j)} (u_k^{(j)})^* G^{1/2} , \quad j = 1, \dots, p , \quad (3.2.6)$$

where $G^{1/2}$ is the unique positive definite square root of G . The verification that these are orthogonal projections is simple.

- That $P_j^* = P_j$ is obvious, since $G^{1/2}$ is positive definite.
- To show that $P_j P_j = P_j$ is also a direct verification using that the matrix U is G -unitary.

To define G -orthogonal projections onto a subspace $\mathcal{V}$ of dimension q (e.g., onto one of the $U^{(j)}$, $j = 1, \dots, p$, above), we resort to the following (and see Exercise 3.2.73-(3)). Let $U \in \mathbb{R}^{n \times n}$ be a G -orthogonal matrix, partitioned as $U = [V, W]$, where $V \in \mathbb{R}^{n \times q}$ and the columns of V span $\mathcal{V}$. Then, we define the G -orthogonal projection onto $\mathcal{V}$ to be

$$P = VV^T G. \quad (3.2.7)$$

Note that $P^2 = P$, but P is not symmetric, at least not in the usual sense; this is actually a deep fact, since in a way “symmetry” is an inner-product dependent property³. What we have now is that $GP = (GP)^T$, or $P = G^{-1}P^T G$, which we may call “symmetry with respect to the G -inner product.”

Also, note that in the case of $\mathcal{V} = U^{(j)}$, for some $j = 1, \dots, p$, then we have $P = G^{-1/2}P_j G^{1/2}$, with P_j as in (3.2.6).

Exercises 3.2.73

- (1) Find the analogous formulas to the “resolution of the identity” and “spectral resolution of A ”, according to the projections in (3.2.6). Do it also for the G -orthogonal projections given by (3.2.7), that is $G^{-1/2}P_j G^{1/2}$.
- (2) Let P be a projection with respect to the G -inner product, as in (3.2.7). What are the eigenvalues of the projection P ? Also, define the complementary projection and determine its eigenvalues.
- (3) Let $\mathcal{V}$ be a subspace of $\mathbb{C}^n$, and let $b \in \mathbb{C}^n$ be given. Find a closed form expression for the solution of the problem

$$\min_{x \in \mathcal{V}} \|x - b\|_G.$$

[Hint: The answer explains that the G -projection in (3.2.7) gives the best approximation with respect to the G -inner product.]

- (4) We are in $\mathbb{R}^3$. Let V be the line of direction $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, and let $G = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 25 \end{pmatrix}$.

³Writing $A = A^T$ really says that $e_i^T A e_j = e_j^T A e_i$.

(a) Find a G -orthogonal matrix whose first column spans V .

(b) Solve the problem $\min_{x \in V} \|x - b\|_G$, where $b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, and find the residual $\|x - b\|_G$ with respect to the optimal solution you found. Compare this solution to the standard least squares solution of this problem.

Exercise 3.2.74 (Symmetric-definite pencil) Let $A \in \mathbb{C}^{n \times n}$ be Hermitian, and $B \in \mathbb{C}^{n \times n}$ be positive definite. Consider the problem of finding the pair (μ, v) , $\mu \in \mathbb{C}$, $v \in \mathbb{C}^n$, $v \neq 0$, such that $Av = \mu Bv$. This is called a generalized eigenvalue problem, μ is called eigenvalue, and v is the associated eigenvector. The pair (A, B) is called pencil, and the specific case considered here is called a symmetric-definite pencil.

- (i) Show that all eigenvalues of a symmetric-definite pencil are real.
- (ii) Show that a symmetric-definite pencil has n linearly independent eigenvectors, and that it can be diagonalized by congruence (that is, both A and B can).
- (iii) Discuss with respect to which matrix G , in the G -inner product, the associated matrix of eigenvectors can be taken to be G -unitary.
- (iv) Finally, assume that all eigenvalues are distinct, and discuss the degree of uniqueness of this G -unitary matrix of eigenvectors. [Hint: See Exercise 3.2.72].

